

Team Worksheet — Lab 23

Full Factorial Design of Experiments (Black Belt Only)

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-23-full-factorial-doe.xlsx`.

3. What we produced

Deliverable: A designed 2^k full factorial experiment with main effects and interaction analysis.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Understand why OFAT fails: it cannot detect interactions at all, and it requires more runs for less information than a factorial design.

Step 2. Recap the DOE vocabulary in full: factor, level, response, run, effect, interaction, replication, randomisation, blocking and experimental error.

Step 3. Select 3 factors from your capstone that you can actually control and vary — temperature, speed, pressure, batch size, staffing level or method variant.

Step 4. Set two levels for each factor — low (-1) and high (+1). Set them wide enough to produce a detectable effect but within safe operating limits.

Step 5. Build the 2^3 design matrix: 8 runs covering every combination of the three factors, in standard order.

Step 6. Decide replication. Replicates give an estimate of pure experimental error and let you test effect significance — 2 replicates gives 16 runs.

Step 7. RANDOMISE the run order. Randomisation protects against unknown time-varying nuisance factors and is not optional.

Step 8. Use blocking if runs must span shifts, days or material lots — block on the nuisance factor so it does not contaminate the effects.

Step 9. Execute the experiment, holding all non-experimental factors constant and recording anything unexpected during each run.

Step 10. Calculate main effects: the average response at the high level minus the average at the low level, for each factor.

Step 11. Calculate interaction effects for AB, AC, BC and the three-way ABC.

Step 12. Plot main effects (steeper slope means larger effect) and interaction plots (non-parallel lines mean interaction present).

Step 13. Test effect significance using a Pareto of effects or a normal probability plot of effects — points off the line are real effects.

Step 14. Build the prediction equation from the significant effects and identify the factor settings that optimise your Y.

Step 15. Run confirmation trials at the predicted optimum. If the confirmation fails, the model is missing something — do not roll out.

7. Self-check

Your design is fully randomised with replication, you have computed and plotted both main effects and interactions, effect significance is assessed statistically, and confirmation runs validate the predicted optimum.

Complete? ☐ Yes ☐ Not yet

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